THE COEFFICIENT AUGMENTATION MAP ANNIHILATES COMMUTATORS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The commutator quotient Lambda^natural is Lambda modulo the additive subgroup generated by commutators, so any element of the form ab - ba maps to zero. invoked at proof line 1644. Makes the final sum vanish in Lambda^natural.

Lemma 0.2. *Let $e \in G$ denote the neutral element. The map*

$$\begin{array}{ccc} \Lambda[G] & \longrightarrow & \Lambda^\natural \\ \sum \lambda_g \cdot g & \longmapsto & \lambda_e \end{array}$$

factors through $\Lambda[G]^\natural$. We denote $\varepsilon : \Lambda[G]^\natural \to \Lambda^\natural$ the induced map.

Proof. We have to show the map annihilates commutators. One has

$$\left(\sum \lambda_g g\right)\left(\sum \mu_g g\right) - \left(\sum \mu_g g\right)\left(\sum \lambda_g g\right) = \sum_g \left(\sum_{g_1 g_2 = g} \lambda_{g_1}\mu_{g_2} - \mu_{g_1}\lambda_{g_2}\right) g$$

The coefficient of e is

$$\sum_g \left(\lambda_g \mu_{g^{-1}} - \mu_g \lambda_{g^{-1}}\right) = \sum_g \left(\lambda_g \mu_{g^{-1}} - \mu_{g^{-1}} \lambda_g\right)$$

which is a sum of commutators, hence it zero in $\Lambda^\natural$. □

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